

* Battery:-

A battery can be defined as the collection of one or more electrochemical cells that convert stored chemical energy into electrical energy. Batteries were introduced to store generated electrical energy. It is important to store energy because it could be used when the generation is down or to power up stand-alone devices that cannot be tethered to the supply from the main's primary.

* Types of Batteries ← Secondary

* Classification of Secondary Batteries:-

- ① Sealed Maintenance - free battery.
- ② lead Acid Battery.
- ③ Lithium Battery.
- ④ Modular Battery.

* SMF Battery (Sealed maintenance free Battery)

The battery requires ~~no~~ minimum maintenance and can be used in supplying consistent and low power such as in UPS applications. These are used to ~~store~~ deliver instant power are used in power deficit.

Areas such as survival areas. These are used with appliances when you need to store important information when the power goes out.

• SMF Battery: -

SMF Battery is / also / known as lead acid battery.

- SMF Battery is also known as VRLA (valve regulated Lead acid Battery). In this type of battery also flat plates are used like lead Acid battery. There is no need to level these batteries again and again with distilled water. This battery is completely shielded, due to which there is no possibility of acid leakage because its structure is of this type.

★ Working: -

These batteries do not require acid and water top up. There is no shortage of water during charging and discharging. In these batteries O_2 is absorbed from the positive plate without taking it out by the Neg plate which is oxygen (recombination principle).

★ Lithium (Li) Battery:-

We all use it in portable devices such as cell phone, a laptop computer, or a power tool. The Lithium battery has been one of the greatest achievements in portable power in the last decade. With use of lithium batteries we have been able to shift from black and white mobile to colour mobiles with additional features like GPS email alert etc. These are the high energy density potential devices for higher capacities. And relatively low self discharge batteries also special cells can provide very high currents to applications such as power tools.

Eg - Torch-light

★ What is the modular battery?

Modular batteries are advanced energy solutions that can be connected in parallel or series to increase capacity or voltage. They are widely used for devices that require high capacity.

or high voltage in order to operate. Before we go ahead and start the modular battery discussion, we'll need to cover a few battery terms first Cells, battery packs and modular batteries. Although these words are usually collectively referred to as 'batteries' each word actually has a different meaning.

A cell is the building block for batteries. One battery cell contains five different parts: the electrolyte, the cathode, the anode, the separator and the housing.

A good example could be the 18650 cylindrical cells or a pouch cell.

A battery pack on the other hand is a finished product, consisting of one or more cells that are connected in series or in parallel with a positive or neg(-ve) terminal for charging and discharging. A battery pack may also contain a PCB (printed circuit board). It is also known as a PCM (protection circuit module) or BMS (Battery management system). a cooling system and possibly other type of protection.

Systems such as hard Gase, multi-series and multi-parallel battery pack certain applications such as electric vehicles and mass power storage - like Solar - grid's require a huge battery pack. To get these we have to create modular batteries. A modular battery is a battery pack that has been designed to work in tandem with other battery packs of the same specification. By introducing or reducing batteries in a modular set up, you'll be able to fulfill your power requirement without being limited to a set capacity voltage. The main advantage of a modular battery is that it is easy to replace. If one of the modules fails you can directly replace that module instead of scrapping the entire battery system. -- failure of one module will not impact the system or the site operation.

Another great advantage of modular batteries is the heat dissipation.

with a good battery management system modular batteries will dissipate heat much better than a single large battery pack.

An example of modular battery ~~can~~ is the Support 12.8 V, 7AH version designed by Cosipow. This modular

This modular battery can support up to 4 groups of batteries in series and 10 groups of batteries in parallel. you can configure the modular battery A/A to the needs of the electrical equipment that you have. If your ~~current~~ previous lead-Acid battery system is 240V, 20AH, battery, then you can use 6-groups of 12.8 V 7AH modular batteries in series and parallel to achieve 256V, 21AH.

Battery

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* Lead Acid Battery : —

Lead Acid batteries are widely used in automobiles invertors, backup power system etc. Unlike tubular and maintenance free batteries, lead acid batteries require proper care and maintenance to prolong its life. The lead acid battery consists of a series of plates kept immersed in sulphuric acid solution. The plates have grids on which the active material is attached. The plates are divided into positive and negative plates. The positive plates hold pure lead as the active material while lead oxide is attached. The plates are divided into positive and negative plates. The positive plates hold pure lead as the active material while lead oxide is attached on the negative plates.

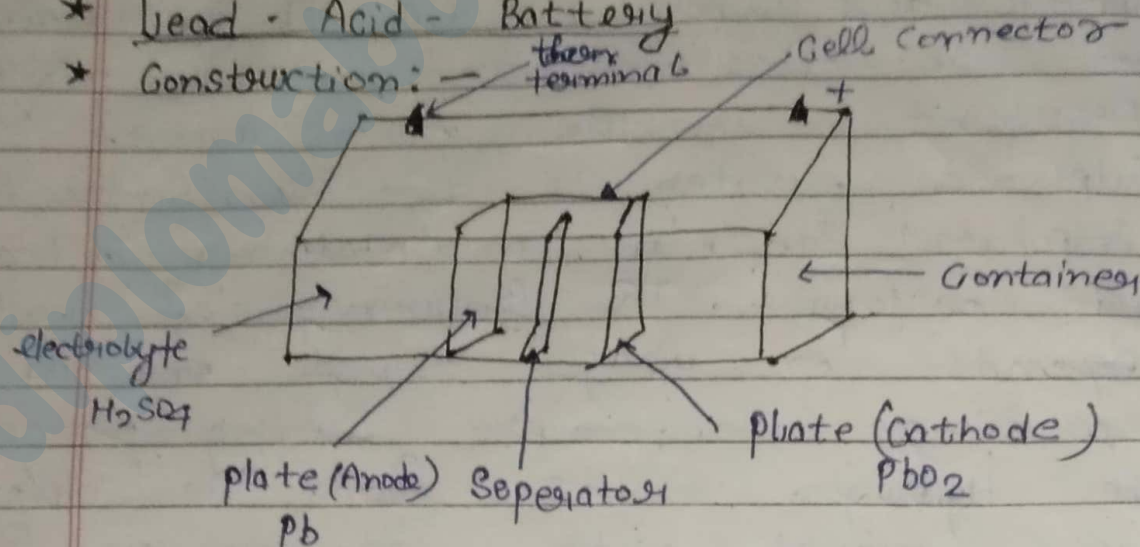
- A complete charged battery can discharge its current when connected to a load, during the process of discharge the

Sulphuric acid combines with the active material on the positive and negative plates resulting in the formation of lead sulphate. Water is the single most important step in maintaining a lead acid battery. The frequency of water depends on usage, charge method and operating temperature. During process the hydrogen atoms from the sulphuric acid react with oxygen to form water.

The results in the release of e^- from the positive plates which will be accepted by the negative plates. The lead to the formation of an electric potential across the battery. The electrolyte in the lead acid battery is a mixture of sulphuric acid and H_2O which can be specific gravity. Specific gravity is the weight of the acid water mixture of sulphuric acid and water which has a specific gravity. Specific gravity is the weight of the acid water mixture compared to equal volume of water. The specific gravity of pure ions free water is 1.

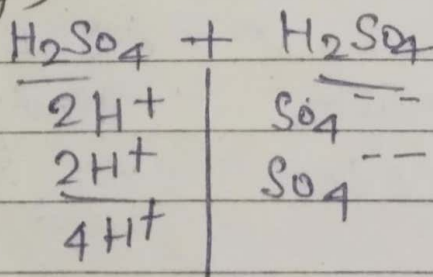
* Lead - Acid - Battery

* Construction:



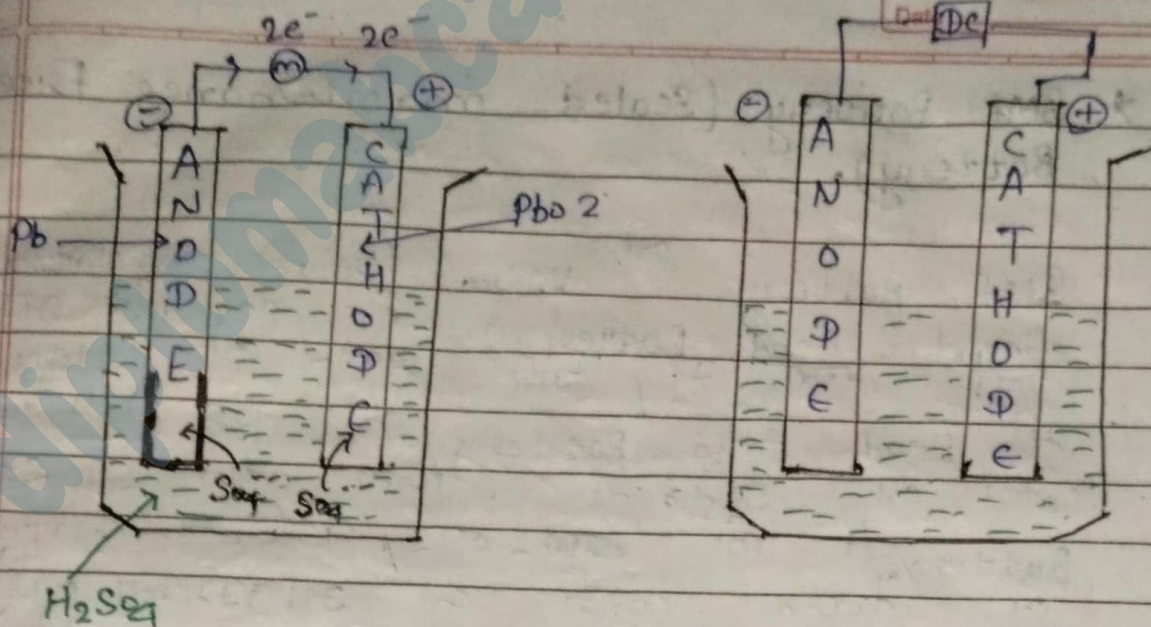
- outer body of battery made with hard rubber or plastic materials.
- plate of anode made with (Pb) material.
- plate of cathode made with PbO_2 material.
- decrease the internal resistance between to plates (electrodes).
- Electrolyte is dilute H_2SO_4 .
- Used to connect the cells in series in battery.
- used for charging and discharging of battery.

(Charging)

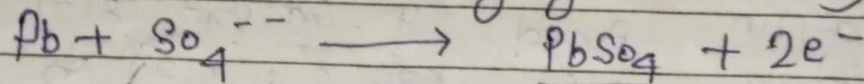


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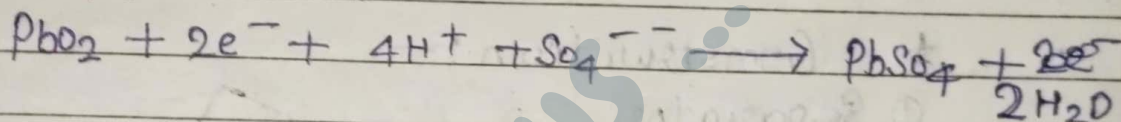


* At Anode:— (Discharging reaction)



* (White precipitate)

At Cathode:—

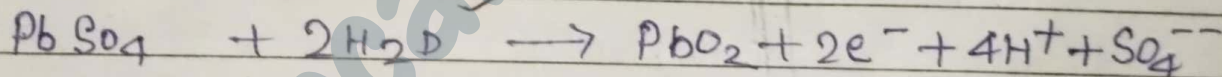


* Charging reaction:—

(At Anode)



(At Cathode)



* Uses:—

In Car

★ Types of Battery: — (Primary batteries)

- Based on applications the type of a battery is chosen.
- For eg — primary batteries ~~and~~ are used in portable devices which run on very low currents such as electronic watch, torch and radio.
- Primary battery are used for one time ^{only} battery.
- They are cheap and easily disposable.
- Some application require high current to run device.

• Secondary Batteries: —

- In such cases the Secondary batteries or rechargeable batteries are used which provide high current as they have high capacity and can be recharged a number of times.
- The rechargeable batteries are costly compared to the ^{other} primary batteries but they are cost effective over long time use.

★ How to Select a Battery?

- The selection of a battery is done by seeing the battery parameters.
- The parameter values for voltage

rating current rating capacity rating number of charging-discharging cycles and shelf life differs from battery to battery.

- The values also differ from one manufacturer to another.
- While designing a solar PV system comprising batteries, the one problem often faced is a proper choice of battery from many available batteries.

* Voltage and current: —

A battery should be selected w.r.t to the voltage and current requirements of the instrument to which it is connected.

* Temperature requirement: —

- The batteries are designed to operate at certain temperature, normally about 25°C .
- In practice the prevailing temp. or ambient temperature may be different.
- Sometimes, the ambient temperature can be above 45°C and sometimes it can be negative or at freezing temperature.
- The battery output current and its capacity depends on the temperature.

- At higher temperature, batteries conduct higher current as compared to lower temperature.

★ Charge - discharge Cycle: -
(If Rechargeable)

- The number of charge - discharge cycle of battery depends on the type of battery.
- Normally, for rechargeable applications, battery with large ^{Number} ~~amount~~ of charge discharge cycle should be chosen.
- For instance, for Standalone solar PV systems, batteries with large number of charge - discharge cycles are chosen.

★ Shelf-life: -

- Batteries shelf life is directly affected by rate of self-discharge.
- Self discharge occurs all the time even when the battery is not used.
- Some batteries have extremely low self discharge rates such as most of the lithium battery whereas some nickel metal hydride batteries can lose up to 4% of their capacity every day.

* Cost: -

- Costing is based on the type of batteries selected and its pricing by different manufacturers.
- Also, it depends on the features of the battery.
- Generally, the maintenance free and advanced technology batteries are expensive.
- In some cases, it may require that the system must perform in all conditions, therefore, the oversizing of the battery is done.

* Availability: -

- Before selecting a battery, we should check its availability in the market.
- If a battery rarely available is used in a system, it may create problems while replacing or buying a new one for the system.
- So, a battery that is readily, easy in transportation and can be bought even in remote places should be selected.

* Ampere hour (Ah or amp hour)

What is an ampere hour (Ah or amp hour)?

Ampere hours - Sometimes abbreviated as AH or amp hours - is the amount of

Ans. Selection Criteria of Batteries for Different Applications

One of the main problems hindering technology revolutions like IoT is power, battery life affects the successful deployment of devices that require long battery life and even though several power management techniques are being adopted to make the battery last longer, a compatible battery must still be selected to achieve the desired outcome. (ii)

Below are some factors to consider when selecting the right type of battery for your project.

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(i) Ampere-Hour Capacity

(ii) Battery Efficiency

Ans.(i) **Ampere-Hour Capacity** : The energy stored in a battery, called the battery capacity, is measured in either watt-hours (Wh), kilowatt-hours (kWh), or ampere-hours (Ahr). The most common measure of battery capacity is Ah, defined as the number of hours for which a battery can provide a current equal to the discharge rate at the nominal voltage of the battery. The unit of Ah is commonly used when working with battery systems as the battery voltage will vary throughout the charging or discharging cycle. The Wh capacity can be approximated from the Ahr capacity by multiplying the AH capacity by the nominal (or, if known, time average) battery voltage. A more accurate approach takes into account the variation of voltage by integrating the AH capacity $\times V(t)$ over the time of the charging cycle.

For example, a 12 volt battery with a capacity of 500 Ah battery allows energy storage of approximately $100 \text{ Ah} \times 12 \text{ V} = 1,200 \text{ Wh}$ or 1.2 KWh. However, because of the large impact from charging rates or temperatures, for practical or accurate analysis, additional information about the variation of battery capacity is provided by battery manufacturers.

(ii) **Battery Efficiency** : The battery efficiency is the ratio of the energy retrieved from the battery, to the energy provided to the battery.

Battery efficiency = Coulombic efficiency $\times$ Ohmic efficiency $\times$ Temperature efficiency.

3	Select and test the battery for a given application	Battery • Types of batteries (Lead acid battery, lithium, sealed maintenance free (SMF) battery, Modular battery). • Selection criteria of batteries for different applications. • Ampere-Hour Capacity. • Efficiency	Testing Condition of a Lead-acid battery	1:0:2
	Select the size of the UPS for a	UPS	Sizing of UPS	1:0:2